

Response to Request for Information:
**Accelerating the Adoption and Use of
Artificial Intelligence as Part of Clinical
Care**

RIN 0955-AA13

**A Framework for Evaluating AI Governance Claims:
Distinguishing Deterministic Governance from Marketing**

Submitted by:

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Deterministic AI Governance Infrastructure

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December 2025

Executive Summary

FERZ AI welcomes the opportunity to respond to HHS's Request for Information on accelerating AI adoption in clinical care. This response serves two purposes: first, to provide concrete recommendations addressing HHS's specific questions; second, and perhaps more importantly, **to equip federal reviewers with a technical framework for evaluating AI governance claims they will inevitably receive from vendors.**

The AI governance market includes many solutions marketed as "guardrails," "responsible AI," "ethical AI," and "trustworthy AI." These terms lack precise technical definitions, making it difficult for federal reviewers to evaluate competing claims. Acquisition officers benefit from distinguishing between:

- **Marketing claims** ("our AI is safe and responsible")
- **Statistical governance** (probabilistic approaches providing confidence intervals)
- **Deterministic governance** (fail-closed enforcement with reproducible, auditable decision artifacts)

This document provides: (1) a taxonomy of AI governance approaches with their respective strengths and limitations; (2) evaluation criteria federal reviewers can use to assess vendor claims; (3) specific questions to ask vendors claiming governance capabilities; and (4) FERZ's recommendations for regulatory frameworks that would accelerate trustworthy AI adoption.

Key Insight: The fundamental barrier to AI adoption in clinical care is not AI capability—it is the inability to *prove* that an AI system's governance was functioning correctly at the time of any specific decision. Without such proof, liability cannot be allocated, regulatory compliance cannot be demonstrated, and patient trust cannot be justified. This is a governance problem, not an AI problem.

Part I: Understanding AI Governance—A Taxonomy for Federal Reviewers

Before addressing HHS's specific questions, we provide a framework for understanding the AI governance landscape. Federal reviewers evaluating vendor claims need this foundation to distinguish substance from marketing.

1.1 The Three Layers of AI Systems

Understanding where governance operates is essential to evaluating vendor claims. AI systems can be conceptualized in three layers:

Layer	Description	Governance Implications
Model Layer	The AI model itself (e.g., GPT-4, Claude, Gemini). Includes training data, architecture, learned parameters.	Model-level governance (RLHF, Constitutional AI) attempts to make the model "safer" through training. Inherently probabilistic—the model <i>tends</i> to behave well but cannot guarantee it.
Application Layer	The software that wraps the model—user interface, integration points, business logic. Where prompts are constructed and outputs processed.	Application-level governance (prompt engineering, output filtering) adds checks around the model. Still largely probabilistic—filters can miss edge cases.
Deployment Layer	The operational environment where AI actions occur—clinical workflows, EHR integration, decision execution. The "last mile" before real-world impact.	Deployment-level governance (FERZ's focus) enforces constraints at the execution boundary. Can be deterministic—nothing executes unless governance validates it.

Critical distinction: Most AI "safety" approaches operate at the Model Layer or Application Layer, where governance is inherently probabilistic. Deterministic governance operates at the Deployment Layer, where execution can be gated by mathematically verifiable constraints.

1.2 The Fundamental Problem: Probabilistic vs. Deterministic Governance

The core technical distinction federal reviewers must understand is between **probabilistic governance** (statistical confidence that a system usually behaves correctly) and **deterministic governance** (mathematical proof that a specific decision complied with specific rules).

Probabilistic Governance: "Likely Compliant"

Probabilistic approaches—including RLHF (Reinforcement Learning from Human Feedback), Constitutional AI, content filtering, and most "guardrails"—operate on statistical principles. They provide confidence that a system *usually* behaves correctly, expressed as accuracy percentages or confidence intervals.

- **How it works:** Train the model to prefer "good" outputs; use classifiers to flag "bad" outputs; filter problematic content before delivery.
- **What vendors claim:** "95% accuracy," "reduced harmful outputs by 80%," "enterprise-grade safety."
- **Risk exposure:** At clinical scale, even small residual failure rates produce frequent governance escapes. The question is whether escapes are (a) prevented pre-execution, and (b) provable/auditable when they occur. Probabilistic approaches cannot guarantee either.
- **Key limitation:** Cannot provide proof that governance was functioning correctly for any specific decision. After an adverse event, there is no way to demonstrate that the system was operating within its governance boundaries.

Deterministic Governance: "Provably Compliant"

Deterministic governance provides fail-closed enforcement with reproducible, independently verifiable audit artifacts *at the governance layer*. This is distinct from claiming the underlying AI model is deterministic—it is not. Determinism is achieved at the **execution boundary** through: (1) pre-execution validation against explicit rules; (2) recorded snapshots of all inputs, policies, and outputs; and (3) reproducible evaluation that can be independently verified.

- **How it works:** Every AI output passes through a validation layer that checks explicit rules. If validation fails, the output is blocked ("deny-by-default" or "fail-closed"). Every decision generates a cryptographic proof bundle capturing inputs, policy state, and verdict.
- **What this enables:** Bit-for-bit reproducible governance decisions, cryptographic audit trails, and bounded residual errors quantifiable by rule coverage and implementation factors.
- **What this means:** In a healthcare deployment, every AI decision is validated against explicit rules before execution. Failed validations are blocked. Every decision can be independently verified and replayed.
- **Key advantage:** After an adverse event, governance compliance can be independently verified by third parties. Liability can be allocated based on provable governance state at decision time.

1.3 Side-by-Side Comparison: Critical Differences

Characteristic	Probabilistic Governance	Deterministic Governance
Compliance assurance	Statistical confidence (e.g., "95% accurate")	Verifiable conformance artifacts with bounded residual risk
Failure mode	"Fail-open": Ungoverned outputs may proceed	"Fail-closed": Nothing executes without validation
After adverse event	Cannot prove governance was working	Cryptographic proof of governance state
Decision reproducibility	Cannot reproduce specific decisions	Bit-for-bit reproducible ("replay test")
Audit capability	Logs of outputs and classifications	Cryptographic proof chain (Proof-Carrying Decisions)
Liability allocation	Unclear—governance state unknowable	Clear—governance state provable
FDA 21 CFR Part 11	Difficult to satisfy—records not verifiable	Designed for compliance—immutable audit trails
Regulatory inspection	Requires trust in vendor's claims	Third-party verification possible

1.4 Why This Distinction Matters for Healthcare

Healthcare is a *precision-critical domain* where statistical assurances are insufficient. Consider:

Scenario: A clinical AI system assists with medication recommendations. The vendor claims high accuracy in avoiding dangerous drug interactions. A large hospital processes thousands of AI-assisted medication decisions monthly.

- **With probabilistic governance:** Some fraction of decisions may have ungoverned drug interaction risks. The fraction may be small, but at scale the absolute number becomes significant. Critically, after an adverse event, there is no mechanism to prove whether governance was functioning correctly for that specific decision.
- **With deterministic governance:** Every medication recommendation is validated against drug interaction rules before reaching the clinician. Failed validations are blocked and escalated. After any decision, cryptographic proof demonstrates exactly which rules were applied and whether they passed.

The difference is not merely technical—it fundamentally changes liability allocation, regulatory compliance posture, and patient safety assurance.

1.5 Emerging Architectures: RAG, Agentic AI, and Governance Topology

Two architectural patterns are accelerating AI deployment in healthcare: **Retrieval-Augmented Generation (RAG)**, which grounds AI outputs in retrieved documents, and **agentic AI**, which enables AI systems to take actions autonomously. Federal reviewers will encounter many proposals featuring these patterns. Understanding their governance implications is essential.

Why RAG and Agentic Patterns Increase Governance Requirements

These patterns do not reduce governance complexity—they increase it:

- **RAG systems** introduce dynamic, unpredictable inputs. The AI's outputs depend on retrieved context that changes with each query—patient records, clinical literature, formulary data. Model-level governance cannot anticipate what documents will be retrieved. Governance must validate outputs regardless of retrieval context.
- **Agentic systems** execute actions with real-world consequences: API calls, database writes, order placements, message transmissions. Post-hoc monitoring cannot reverse an incorrectly administered medication, a misconfigured treatment plan, or an erroneous prior authorization denial. Governance must operate pre-execution, not post-detection.

Reviewer Note: Federal reviewers should be skeptical of claims that RAG or agentic architectures are inherently "safer" or "more accurate." These architectures shift *where* errors occur, not *whether* they occur. The fundamental question remains: can you prove governance was functioning for any specific decision?

Governance Topology in Multi-Agent Systems

As agentic AI evolves toward multi-agent architectures—where specialized agents collaborate on complex tasks—a critical architectural question emerges: **where does governance operate?**

Per-Agent Governance distributes guardrails across individual agents. This creates structural vulnerabilities:

- Agent-to-agent communication may bypass governance entirely
- Emergent behaviors from agent interactions remain ungoverned
- No unified audit trail across the agent ensemble
- Inconsistent policy enforcement across agents with different guardrail implementations
- "*Governance arbitrage*"—sophisticated actors may route sensitive operations through the least-governed agent

Centralized Governance at the Execution Boundary validates all actions—regardless of which agent originates them—before they execute. This architecture ensures consistent policy enforcement, unified auditability, and captures emergent behaviors at the point of real-world impact. The governance layer is model-agnostic and agent-agnostic; it governs *actions*, not *agents*.

Example: Consider a clinical agentic system with a retrieval agent (pulls patient records, literature), a reasoning agent (synthesizes recommendations), and an action agent (places orders, sends messages). With per-agent governance, each agent has separate guardrails optimized for its function—but no single point validates that the *chain* of agent interactions produced a compliant outcome. With centralized execution-boundary governance, every action passes through deterministic validation regardless of which agent initiated it, and the audit trail captures the full decision chain.

Questions for Multi-Agent AI Proposals

Federal reviewers evaluating multi-agent AI proposals should ask:

1. Does governance operate per-agent or at the execution boundary?
2. If per-agent, how are inter-agent communications governed?
3. How are emergent behaviors from agent interactions detected and controlled?
4. Is there a unified audit trail across all agents, or separate logs per agent?
5. Can you replay a decision that involved multiple agents and obtain identical results?

Part II: Evaluating AI Governance Claims—A Guide for Federal Reviewers

Federal acquisition officers will receive numerous proposals claiming AI governance capabilities. This section provides practical tools to evaluate those claims.

2.1 Red Flags: Marketing Language vs. Technical Substance

The following table identifies common marketing claims and provides questions that distinguish substance from noise:

Vendor Claim	What It Usually Means	Questions to Ask
"AI Guardrails"	Content filtering or output classifiers. Probabilistic. May miss edge cases.	What happens when guardrails fail? Can you prove they were active for a specific decision?
"Responsible AI"	Training-time interventions (RLHF, Constitutional AI). Model tends to behave responsibly.	How do you verify responsible behavior for a specific decision? What are your error bounds?
"Enterprise-Grade Safety"	Marketing term. No technical definition. May mean access controls or logging.	Define "enterprise-grade." What specific guarantees do you provide? How are they verified?
"Explainable AI"	Post-hoc rationales. May explain why AI probably made a decision, but cannot prove compliance.	Can explanations be independently verified? Can decisions be replayed to confirm the explanation?
"AI Monitoring"	After-the-fact analysis. Detects problems after they occur, not before.	How do you prevent violations rather than detect them? What is your pre-execution validation?
"99% Accurate"	1% of decisions are ungoverned. At scale, this means thousands of potential failures.	What happens to the 1%? Can you identify which specific decisions are in that 1%?

2.2 The Four Tests: Evaluating Governance Capabilities

FERZ has published the **Four Tests Standard (4TS)** as a vendor-neutral framework for evaluating AI governance claims. Federal reviewers can use these tests to assess any governance solution:

Test	Question It Answers	How to Verify
Stop Test	Can the system be halted before side-effects occur? Does governance validation happen before execution, not after?	Ask: "What happens if governance validation fails?" The answer must be "the action is blocked," not "we log it for review."

Test	Question It Answers	How to Verify
Ownership Test	Is there clear authority signing governance policy? Can you identify who authorized the AI's operational boundaries?	Ask: "Show me the signature chain for governance policy." There must be cryptographic proof of who authorized what.
Replay Test	Can any historical decision be reproduced and verified? Can a regulator independently confirm what happened?	Ask: "Show me a decision from last week and reproduce it." The replayed decision must match the original exactly.
Escalation Test	Are human oversight routes defined? When governance denies an action, what happens? Is there mandatory custody transfer?	Ask: "What happens when you deny an action?" There must be defined escalation paths with human decision-makers.

Key insight: A governance system that cannot pass all four tests provides statistical confidence at best. It cannot provide the proof-level assurance required for regulated healthcare environments.

2.3 Technical Due Diligence Questions

When evaluating AI governance solutions for healthcare, federal reviewers should ask these specific technical questions:

Architecture Questions

6. At what layer does your governance operate? (Model, Application, or Deployment)
7. Is governance validation performed before or after execution?
8. What is the default behavior when governance cannot validate an action? (Fail-open vs. fail-closed)
9. Can your system operate with any AI model, or is it tied to a specific model provider?

Compliance Verification Questions

1. How do you prove that governance was functioning correctly for a specific historical decision?
2. Can a third party independently verify your compliance claims without trusting your attestations?
3. What cryptographic mechanisms ensure audit trail integrity?
4. How do you satisfy FDA 21 CFR Part 11 requirements for electronic records and signatures?

Error Bound Questions

1. What are your quantified error bounds? (Not accuracy percentages—actual mathematical error bounds)
2. How did you derive those bounds? (Empirical testing vs. mathematical proof)
3. What assumptions must hold for those bounds to be valid?
4. How do you detect and handle cases that fall outside your bounded domain?

Operational Questions

1. What is your governance validation latency? (Must not significantly impact clinical workflows)
2. How do you handle governance policy updates without disrupting operations?
3. What happens if your governance system experiences an outage?
4. Can you demonstrate air-gapped deployment for sensitive environments?

Part III: Responses to HHS Specific Questions

Having established the technical framework, we now address HHS's specific questions from the perspective of deterministic governance.

Question 1: Barriers to Private Sector Innovation

"What are the biggest barriers to private sector innovation in AI for health care?"

The primary barriers are governance-related, not capability-related:

1. **The Governance Gap:** Healthcare requires certainty; current AI governance provides only statistical confidence. Organizations cannot deploy AI when they cannot prove compliance.
2. **Liability Uncertainty:** Without clear governance boundaries, liability cannot be allocated. Legal departments block deployments when governance state is unknowable.
3. **Audit Trail Inadequacy:** FDA 21 CFR Part 11 requires verifiable electronic records. Current AI systems produce outputs, not evidence. There is no cryptographic proof that governance was functioning at decision time.
4. **Regulatory Uncertainty:** Non-medical device AI falls into gray zones. Organizations cannot invest when regulatory requirements are unclear.

FERZ Recommendation: HHS should establish clear governance requirements that distinguish between statistical and deterministic assurance levels, creating a pathway for organizations to deploy AI with appropriate governance for their risk tolerance.

Question 2: Regulatory and Policy Changes

"What regulatory, payment policy, or programmatic design changes should HHS prioritize?"

Regulatory Recommendations with CFR Citations

Note: The following citations are illustrative, not exhaustive. We recommend HHS issue guidance clarifying application to AI governance rather than amending the underlying regulations.

1. **21 CFR Part 11 (Electronic Records and Signatures):** Issue guidance recognizing cryptographic proof bundles (e.g., Proof-Carrying Decisions) as satisfying §11.10 requirements for audit trails and §11.50 requirements for signature manifestations. Current language contemplates traditional electronic signatures; guidance should clarify that cryptographic attestation of policy state satisfies the integrity and attribution requirements.
2. **45 CFR Part 170 (Health IT Certification):** Consider adding AI governance attestation requirements to §170.315 certification criteria. Systems claiming clinical decision support functionality should demonstrate reproducible governance validation as a certification condition.
3. **42 CFR Part 493 (CLIA):** Issue guidance on how AI-assisted laboratory analytics should satisfy §493.1251 procedure manual requirements and §493.1283 retention requirements. Deterministic governance artifacts could satisfy the "complete record" requirements.
4. **45 CFR Part 164 (HIPAA Security):** Clarify how AI governance audit trails interact with §164.312(b) audit control requirements and §164.312(c) integrity

requirements. Cryptographic proof chains may exceed current compliance expectations; guidance should recognize this.

Payment Policy Recommendations

- **Governance Attestation as Reimbursement Condition:** For AI-assisted clinical workflows in high-risk categories, require governance attestation artifacts as a condition of reimbursement. This creates market incentive for governance adoption.
- **Incentives for Burden-Reduction Tools:** Provide enhanced reimbursement for AI-assisted documentation, coding, and prior authorization when governance artifacts enable post-payment integrity verification. This aligns payer interests with governance adoption.
- **Pilot Reimbursement Models:** CMS should pilot reimbursement models for AI tooling where governance artifacts support audit. This generates evidence on governance value while managing risk.

Question 3: Novel Legal Issues

"What novel legal and implementation issues exist for non-medical devices?"

1. **The Black Box Problem:** Current AI cannot prove governance state at decision time. This creates fundamental liability allocation challenges. Who is responsible when governance cannot be verified?
2. **Temporal Governance:** AI systems may be updated between deployment and adverse event. Without governance infrastructure capturing policy state at decision time, retrospective analysis is impossible.
3. **Multi-Party Accountability:** Clinical AI involves model developers, governance providers, deploying institutions, and clinicians. Clear boundary definitions are needed.
4. **Privacy-Preserving Audit:** HIPAA requires protecting PHI while maintaining auditable records. Deterministic governance with cryptographic proofs can address this, but regulatory recognition is needed.

HHS's Role: Define minimum governance architecture standards for clinical AI; create accountability frameworks for multi-party deployments; publish reference architectures demonstrating acceptable approaches.

Mapping to FDA SaMD and CDS Realities

Although this RFI addresses AI broadly, reviewers will implicitly map responses to existing FDA frameworks. Deterministic governance artifacts support auditability and change control regardless of whether a tool is regulated as a device. For clinical decision support (CDS) that falls outside device regulation under 21st Century Cures Act criteria, governance artifacts nonetheless provide the documentation and traceability that healthcare organizations require for institutional risk management. For tools that do qualify as Software as a Medical Device (SaMD), these same artifacts directly support FDA expectations for traceability, change control, and post-market surveillance. The governance layer thus provides compliance infrastructure that scales across the regulatory spectrum—from unregulated CDS to Class II/III SaMD—without requiring different architectures for different regulatory categories.

Question 4: AI Evaluation Methods

"What are the most promising AI evaluation methods for clinical care?"

The most promising evaluation methods focus on governance verification rather than AI capability assessment:

Pre-Deployment Evaluation

- **Governance Coverage Analysis:** What percentage of AI outputs pass through deterministic validation? Target: 100% for high-risk decisions.
- **Rule Completeness Testing:** Do governance rules cover all compliance requirements (HIPAA, FDA, state regulations)?
- **Adversarial Testing:** Systematic testing of governance behavior under unusual or adversarial inputs.

Post-Deployment Evaluation

- **Replay Verification:** Can any historical decision be replayed and obtain identical results? This enables post-hoc audit without trusting logs.
- **Escalation Analysis:** How often do decisions escalate to human review? Are thresholds appropriately calibrated?
- **Governance Drift Detection:** Monitor for changes in AI behavior that might indicate model drift affecting governance effectiveness.

HHS Opportunity: Support development of governance evaluation tools and conformance testing infrastructure. The 4TS conformance bundle provides a model.

Questions 5-10: Summary Responses

Question 5 (Private Sector Activities): HHS should recognize deterministic governance in certification criteria; fund pilot programs; create testing infrastructure; and establish public-private partnerships for reference architectures.

Question 6 (AI Tool Performance): AI succeeds in domains with clear boundaries and low compliance complexity (imaging, analytics). It underperforms in governance-heavy domains. The common thread is governance inadequacy, not capability limitation. Deterministic governance unlocks high-value applications currently blocked by compliance concerns.

Question 7 (Decision-Makers): General Counsel and Risk Management are typically the blocking stakeholders—they cannot approve deployments without governance guarantees. CMOs focus on clinical efficacy; CIOs on integration; Compliance Officers on regulations. Deterministic governance directly addresses the Risk Management/Legal blockers.

Question 8 (Interoperability): Critical gaps exist in standards for exchanging governance attestations, audit trail formats, and AI governance metadata in FHIR. The 4TS Proof-Carrying Decision (PCD) format provides a foundation. **Concrete recommendation:** HHS could sponsor an HL7 FHIR Implementation Guide for exchanging governance attestations (decision hashes, policy IDs, cryptographic signatures) alongside CDS Hooks responses or as extensions to relevant FHIR resources. This would enable EHRs and clinical systems to verify governance state without proprietary integrations.

Question 9 (Patient Concerns): Patient concerns about privacy, accuracy, dehumanization, and bias are fundamentally governance concerns. Deterministic governance addresses them: cryptographic audit trails ensure verifiable privacy compliance; reproducible decisions enable accuracy demonstration; escalation protocols preserve human oversight; bias detection provides equity assurance.

Question 10 (Research Priorities): HHS should prioritize research in deterministic governance methods, cryptographic audit infrastructure, causal inference for clinical AI, and human-AI collaboration frameworks.

Part IV: About FERZ—Deterministic Governance Infrastructure

FERZ AI develops deterministic AI governance infrastructure software. Unlike approaches that attempt to make AI models "safer" through training, FERZ operates at the deployment layer—wrapping any AI system in a deterministic validation shell that guarantees compliance with explicitly defined rules.

Core Technology

- **LASO(f)**: Multi-tier linguistic and action governance framework. Eight validation tiers (syntax, semantics, pragmatics, stylistics, discourse, morphology, phonology/graphemics, lexicon) with bidirectional propagation. Designed for low-latency, high-accuracy validation.
- **DELIA**: Flat constraint architecture with Z3 SMT solver for rapid deterministic validation. Ideal for action governance requiring ALLOW/DENY/ESCALATE decisions.
- **CausaCore**: Cross-domain causal modeling with four specialized engines for drug interaction analysis, clinical trial design, and precision medicine.
- **4TS Standard**: Vendor-neutral open standard for AI governance verification (CC BY-NC-ND 4.0).

Performance benchmarks available upon request under appropriate evaluation conditions; figures depend on policy complexity and deployment environment.

Regulatory Alignment

FERZ systems are designed for regulatory compliance from the ground up:

- **FDA 21 CFR Part 11**: Electronic signatures, immutable audit trails, record integrity verification
- **HIPAA**: Access controls, encryption, audit logging for protected health information
- **EU AI Act**: High-risk system governance, documentation requirements, explainability
- **SEC Rule 17a-4**: WORM storage, tamper-evidence, 7-year retention

Offer of Collaboration

FERZ stands ready to support HHS:

1. **Pilot Programs**: Deploy LASO(f) and DELIA in hospital pilots to generate real-world evidence
2. **Standards Development**: Collaborate on healthcare-specific adaptations of the 4TS Standard
3. **Technical Consultation**: Provide briefings to HHS staff on deterministic governance approaches
4. **Research Partnerships**: Participate in HHS-sponsored research through cooperative agreements or CRADAs

Vendor-Neutral Guidance: We encourage HHS to adopt the evaluation criteria and governance principles described in Parts I and II regardless of vendor. The Four Tests Standard (4TS) is published under open license specifically to enable vendor-

neutral assessment. Our interest is in establishing clear standards that accelerate trustworthy AI adoption—FERZ benefits from a well-defined market, not from regulatory ambiguity.

The promise of AI in clinical care is immense. The barrier is trust. Deterministic governance infrastructure can bridge that gap—transforming AI from a liability concern into a verified asset.

Public Submission Notice: This submission contains no protected health information (PHI) and no confidential business information. It is intended for inclusion in the public docket.

Respectfully submitted,

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